

Math 113 – Abstract Algebra

Homework 3 — my solutions

Fall 2026 ■ Keshav Ramamurthy

All assignments must be **turned in on Gradescope** by the deadline. Late assignments are not accepted.

Have questions? Work with your peers, attend office hours, check out the [Student Learning Center](#), and ask in discussion section!

Problems for credit. Turn in these problems; some subset of them will be graded.

1. Fraleigh Exercise 6.6
2. Fraleigh Exercise 6.14
3. Fraleigh Exercise 6.34
4. Fraleigh Exercise 6.44
5. Fraleigh Exercise 7.4
6. Fraleigh Exercise 7.8
7. Fraleigh Exercise 7.19

Practice problems. The following problems will not be graded, and should not be turned in, but are recommended as good practice problems. Solutions to most odd-numbered exercises are available in the back of the textbook.

- Fraleigh Exercises 6.3, 6.7, 6.9, 6.13, 6.17, 6.33, 6.35, 6.45, 6.47, 6.49, 6.51
- Fraleigh Exercises 7.1, 7.3, 7.7, 7.11, 7.13, 7.15, 7.17

Solutions

Problem 1 (Fraleigh Exercise 6.6)

Solution. $\gcd(48, 88) = 8$ as $48 = 8 \cdot 6$, $88 = 8 \cdot 11$ and 8 and 11 are relatively prime. $\square$

Problem 2 (Fraleigh Exercise 6.14)

Solution. On $\mathbb{Z}_8$, we are examining functions $f : G \rightarrow G$ that satisfies

$$x, y \in G, f(x + y) = f(x) + f(y)$$

Notice for $\mathbb{Z}_8$, $\mathbb{Z} = \langle 1 \rangle$ So we want to find the number of generators $\langle g \rangle$ that can generate $\mathbb{Z}_8$.

Let's examine what other values can be generators. Suppose $G = \langle k \rangle$ for $k \neq 1$.

This would generate $0, k, 2k, \dots$ modulo 8.

This would not generate the entire residue class mod 8 if nk reaches 0

This is because notice if $nk = 0$ has a solution, then $nk \equiv 1 \pmod{8}$ would not have a solution, meaning not all values would be hit.

The residues relatively prime to 8 are 1, 3, 5, 7 so the answer is 4. $\square$

Problem 3 (Fraleigh Exercise 6.34)

Solution. We attempt to find an example of an infinite group that is not cyclic. Every infinite cyclic group must be isomorphic to $\langle \mathbb{Z}, + \rangle$

Consider $G = \{Q \setminus \{0\}, *\}$ Let's first examine if its a group.

It has a identity 1, and notice the inverse for $a \in Q \setminus \{0\}$ is $\frac{1}{a}$.

Closure also holds as $p, q \in Q \setminus \{0\}$, $pq \in Q \setminus \{0\}$

We have to show that this group is not cyclic. For sake of contradiction, suppose there exists a generator for this group $\langle k \rangle$ Consider arbitrarily 2 and 3, we'd have

$$\exists n, m \in \mathbb{Z}^+ \text{ s.t } k^n = 2, k^m = 3$$

Notice, we'd have $(k^n)^m = 2^m$, $(k^m)^n = 3^n \rightarrow 2^m = 3^n$ which is a contradiction as n and m are positive integers (this only holds for $m = n = 0$.) $\square$

Problem 4 (Fraleigh Exercise 6.44)

Solution. We have a cyclic group G with generator a and G' a group such that $G \cong G'$

We have a function $\phi : G \rightarrow G'$. and we wish to show that $\phi(x)$ is completely determined by the value $\phi(a)$. So if two isomorphisms have the same value at a , we wish to show they are the same function.

Notice we can write the inputs as $\phi(a^1) = \varphi(a^1) = b$ for some $b \in G'$

And we wish to basically show that this implies $\phi(a^i) = \varphi(a^i)$ for all i (on $\mathbb{Z}$) We first consider the homomorphism property on ϕ :

$$\phi(x * y) = \phi(a^{i+j}) = \phi(a^i)\phi(a^j)$$

where here $x = a^i, y = a^j$. Notice this creates

$$\phi(a^2) = \phi(a)\phi(a) = \phi(a)^2$$

and similarly $\phi(a^i) = \phi(a)^i$. (and holds for negatives because homomorphism preserves inverses) Notice this applies analogously to φ , so $\varphi(a^i) = \varphi(a)^i$. Thus, we are done, as any

$$\forall x \in G, \exists i \in \mathbb{Z}, a^i = x \rightarrow \phi(a^i) = \varphi(a^i) = \phi(a)^i = \varphi(a)^i \text{ and } \phi(a) = \varphi(a) \text{ as given}$$

so for all values of x , these are the same.

□

Problem 5 (Fraleigh Exercise 7.4)

Solution. The elements of the subgroup generated by $\{12, 30\}$ of $\mathbb{Z}_{36}$

We first consider the values $12k$ attains mod 36, and we can pretty easily see that it goes $0, 12, 24, 0, \dots$

For 30, it goes $0, 30, 24, 18, 12, 6, 0, \dots$

So the subgroup of $\mathbb{Z}_{36}$ is $\{0, 6, 12, 18, 24, 30\}$ or just the subgroup with generator 6.

□

Problem 6 (Fraleigh Exercise 7.8)

Solution. We start at a^2b in the diagram.

Each subsequent multiplication by a we go around following the arrows.

So $(a^2b)a$ is ab , $(a^2b)a^2 = b$, and finally $a^2b(a^3) = a^3b$ which is our answer.

□

Problem 7 (Fraleigh Exercise 7.19)

Solution. We wish to show the existence of a non abelian group of size $2n$ for $n \geq 3$ generated by two elements of order 2.

Consider the first where on $(1, 2, \dots, n)$ you keep 1 as a fixed point and reverse $(2, \dots, n)$

Now consider the second manipulation that reverses the entire permutation, so

$$(1, 2, \dots, n) \rightarrow (n, n-1, n-2, \dots, 1) \rightarrow (1, 2, \dots, n)$$

Consider applying both operations, and examining position one.

Let's track 1 first, then examine the rest. $1 \rightarrow 1 \rightarrow n \rightarrow 2 \rightarrow n-1 \rightarrow 3 \rightarrow n-3 \rightarrow 4 \rightarrow n-4 \dots$

So it seems like it generates all $2n$ and we'll show it rigorously.

Let the first be defined by a function A where $A(1) = 1$ and $A(i) = n+2-i$ for $n \geq i \geq 2$.

And let's define the second as $B(i) = n+1-i$ for $n \geq i \geq 1$ $B(A(1)) = n$.

And, $T = B(A(i))$ for $i \geq 2$ is $B(n+2-i) = (n+1) - (n+2-i) = i-1$. So thus, $1 \rightarrow n \rightarrow n-1 \rightarrow \dots \rightarrow 2$ hits all n elements.

We can generate another n by applying A after all the terms above (which are distinct for the same idea, they enumerate them all in order).

To show that this creates $2n$ distinct elements, we consider the function on 2, and for sake of contradiction assume $T^i A(2) = T^j(2)$

$$A(2) = n \rightarrow T^i(A(2)) = T^{i+1}(n)$$

Now let's manipulate $T^j(2)$. We get $2 = T^{n-1}(1)$ so $T^j(2) = T^{j+n-1}(2)$

$$T^{i+1}(n) = T^{j+n-1}(n)$$

Since powers of T were distinct with order n , setting the exponents equal modulo n yields

$$i+1 \equiv j+n-1 \pmod{n} \rightarrow i-j \equiv -2 \pmod{n}$$

Now, to finish up, let's apply this back onto 1.

On 1, we'd have

$$T^i A(1) = T^j(1) \rightarrow T^i(1) = T^j(1)$$

because $A(1) = 1$, and this yields $i \equiv j \pmod{n}$, which is a contradiction, so there must be $2n$ distinct elements.

Now, let's show that it isn't abelian.

Notice we showed $B \circ A$ has order $n \geq 3$, and for sake of contradiction suppose G is abelian.

Then, we have that $B \circ A = A \circ B$ and $(BA)^2 = BABA = BBAA$ and since these are both order two, this would mean that this would be the identity or permute back in this case, which is a contradiction, as this would imply an order less than 3. $\square$